

Theorem & Proof: Second-Order Linear ODEs

Theorem Statement

Formulations

Let ϕ_1, ϕ_2 be solutions of equation $L(y) = y'' + a_1 y' + a_2 y = 0$ given by:

$$\begin{cases} \phi_1(x) = e^{r_1 x}, \quad \phi_2(x) = e^{r_2 x} & (r_1 \neq r_2) \\ \phi_1(x) = e^{r_1 x}, \quad \phi_2(x) = x e^{r_1 x} & (r_1 = r_2) \end{cases}$$

1. Forward Statement: If C_1, C_2 are constants, then $\phi = C_1 \phi_1 + C_2 \phi_2$ is a solution of $L(y) = 0$ on $-\infty < x < \infty$.

2. Converse Statement: If ϕ is any solution of $L(y) = 0$, there exist **unique constants** C_1, C_2 such that $\phi = C_1 \phi_1 + C_2 \phi_2$.

Proof: Part I

Superposition

Let ϕ_1, ϕ_2 be solutions of $L(y) = 0$. Thus:

$$L(\phi_1) = 0 \quad \text{and} \quad L(\phi_2) = 0$$

Let C_1, C_2 be any constants. Define $\phi = C_1 \phi_1 + C_2 \phi_2$.

Since L is a linear differential operator:

$$L(\phi) = C_1 L(\phi_1) + C_2 L(\phi_2) = 0$$

This proves that ϕ is indeed a solution of $L(y) = 0$ on $-\infty < x < \infty$.

Proof: Part II

Uniqueness

Conversely, let ϕ be any solution of $L(y) = 0$. Choose any real number x_0 and let $\phi(x_0) = \alpha$, $\phi'(x_0) = \beta$.

This implies ϕ solves the Initial Value Problem (IVP):

$$L(y) = 0, \quad y(x_0) = \alpha, \quad y'(x_0) = \beta$$

By the **Existence Theorem**, IVP has a unique solution $\psi = C_1 \phi_1 + C_2 \phi_2$ satisfying $\psi(x_0) = \alpha$, $\psi'(x_0) = \beta$.

By the **Uniqueness Theorem**:

$$\phi = \psi = C_1 \phi_1 + C_2 \phi_2$$

where C_1, C_2 are uniquely determined by α, β . ■